

RESEARCH ARTICLE

Marine protected areas enhance climate resilience to severe marine heatwaves for kelp forests

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Abstract

1. Marine protected areas (MPAs) are an important tool for conserving coastal marine ecosystems, with well-documented benefits for fished species. However, their potential to benefit non-exploited species, such as primary producers in kelp forest ecosystems, is less well understood, particularly under escalating climate change impacts.
2. In this study, we used four decades of remote sensing to examine the effects of 54 MPAs on kelp canopy coverage and assess how these effects influence kelp resilience to marine heatwaves. We developed a method for identifying paired reference (control) sites using historical satellite data and then used Before-After Control-Impact Paired Series analysis to examine whether the implementation of MPAs leads to increases in kelp coverage. In addition to examining changes in kelp coverage before and after MPA implementation, we also analysed the effect of MPAs on the resistance and recovery of kelp canopy coverage to a series of severe marine heatwaves in the North Pacific between 2014 and 2016.
3. We found that the implementation of MPAs led to a modest positive effect with an 8.5% increase in kelp coverage compared to reference areas, though effects varied across MPAs.
4. The positive effect of MPAs became more evident following the marine heatwaves, with kelp forests in MPAs showing greater recovery than in reference sites, particularly in southern California.
5. *Synthesis and applications.* Our results provide empirical evidence of the potential role of MPAs as climate adaptation tools and highlight that well-managed MPAs can support ecosystem stability under increasing climate stress.

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KEYWORDS

30 by 30 targets, Before-After Control-Impact Paired Series, climate disturbance effects, climate-smart protected areas, ecosystem recovery, kelp forests, marine heatwaves, reference site selection

1 | INTRODUCTION

Coastal marine habitats are facing a variety of environmental pressures, including marine heatwaves (MHWs), trophic disruptions and disease outbreaks (Eger et al., 2022; Ling et al., 2015; Rogers-Bennett & Catton, 2019). As a result, there is a need for conservation and restoration tools to mitigate habitat loss, maintain biodiversity and promote the resilience of these systems (Arafah-Dalmou et al., 2023; Bertocci et al., 2015; Caselle et al., 2015; Foster & Schiel, 2010; Hopf et al., 2022). Marine protected areas (MPAs) are effective management tools that enhance populations of marine species, promote ecological integrity and in some cases may provide resilience to climate impacts (Benedetti-Cecchi et al., 2024; Edgar et al., 2014; Grorud-Colvert et al., 2021; Micheli et al., 2012; Olguín-Jacobson et al., 2025).

Marine protection can provide climate resilience in a number of ways, including both by mitigation and adaptation (Jacquemont et al., 2022; Olguín-Jacobson et al., 2025) with potential mechanisms at the organismal, population and ecosystem levels (White et al., 2025). For example, Micheli et al. (2012) showed that MPAs in Mexico had larger individuals of two species of abalone than fished areas and that those larger individuals were less affected by a hypoxia event. At the community level, temperate MPAs in both California (Caselle et al., 2015; Eisaguirre et al., 2020; Kumagai et al., 2024) and Tasmania (Ling et al., 2009) contained larger and more abundant predators than outside the protected areas, with the potential for cascading trophic effects on herbivores and primary producers. Trophic cascades may be particularly important to the maintenance of biogenic habitat, such as kelp and seagrass (Hughes et al., 2013), but there is still some debate (Malakhoff & Miller, 2021; Reed et al., 2011).

When MPAs do provide indirect benefits to primary producers by preserving or strengthening trophic cascades, they may also enhance resilience to other environmental stressors, including those associated with climate change (Kumagai et al., 2024; Ling et al., 2009; Ling & Johnson, 2012; White et al., 2025). While MPAs themselves are unlikely to directly mitigate the effects of increased temperature, they may be able to promote the recovery of biogenic habitat by reducing other stressors such as direct harvest and enhancing ecosystem resilience to abiotic stressors (Caselle et al., 2015; Eisaguirre et al., 2020; Filbee-Dexter et al., 2024; Hamilton et al., 2023; Lester & Halpern, 2008; Teck et al., 2017). Although recent work is revealing diverse mechanisms by which MPAs can provide climate resilience (Benedetti-Cecchi et al., 2024; Kumagai et al., 2024; Micheli et al., 2012; Olguín-Jacobson et al., 2025; Ziegler et al., 2023), there is limited empirical evidence on the benefits that MPAs can provide

against climate impacts at large spatial and temporal scales (Bates et al., 2019).

One powerful way to assess the effects of MPA protection is by collecting data before and after MPA implementation in both control and impact sites (Kerr et al., 2019; Malakhoff & Miller, 2021). Ideally, the control and intervention sites will experience similar environmental conditions and therefore show similar trends and variability prior to the impact (Stevens, 2002; Wauchope et al., 2021). However, obtaining data over the necessary scales of space and time to conduct these analyses can be difficult. There is often a lack of pre-intervention data to inform the selection of reference sites, forcing researchers to rely on proxies, including physical site characteristics (Caselle et al., 2015; Hopf et al., 2022; Wauchope et al., 2021). However, if extensive pre-intervention data can be obtained (e.g. through historical remote sensing), then paired control sites can be identified by finding locations that showed similar dynamics to the MPA sites before the MPA was established.

Kelp forests along the coast of California provide an ideal system in which to examine these issues, as there is an extensive MPA network and these ecosystems have been negatively affected by severe MHWs in recent years. In 1999, the State of California passed the Marine Life Protection Act (MLPA), which mandated the re-design of state-implemented MPAs in California into a cohesive, ecologically connected network (Botsford et al., 2014). By 2012, California had implemented 124 MPAs covering more than 16% of its state waters (Botsford et al., 2014; CDFW & COPC, 2018). Kelp forests are one of the ecosystems targeted for protection by this network. Climatic disturbances, namely the severe 2014–2016 MHWs that impacted the west coast of North America, have led to declines in kelp coverage throughout most of the region (Arafah-Dalmou et al., 2019; Cavanaugh et al., 2019; McPherson et al., 2021; Rogers-Bennett & Catton, 2019; Teck et al., 2017). Simultaneously, sea star wasting disease (SSWD) caused a collapse in sea star populations, which are an important predator of sea urchins that may regulate sea urchin numbers (Duggins, 1983; Galloway et al., 2023; Whippo et al., 2024). The combination of heatwave and grazing stressors led to large declines in kelp abundance (Beas-Luna et al., 2020; Bell et al., 2023; Hewson et al., 2018; McPherson et al., 2021; Rogers-Bennett & Catton, 2019; Smith et al., 2024).

In this study, we examined the effects of MPAs on kelp canopy coverage in California using a 39-year time series of satellite observations of kelp surface canopy from 1984 to 2022. To assess MPA impacts, we first developed a method to identify paired reference sites for each MPA that exhibited similar trends and variability in kelp coverage prior to MPA implementation. We then used BACIPS (Before-After Control-Impact Paired Series) analysis to examine the

effects of MPA protection on kelp coverage at the site level and regionally. We tested (1) whether kelp coverage increased in MPAs relative to reference sites after MPA implementation and (2) whether MPAs provided resistance to and/or recovery from the 2014–2016 MHWs on kelp forests.

2 | MATERIALS AND METHODS

2.1 | Data collection

MPA boundaries were obtained from the California Department of Fish and Wildlife for MPAs along the coast of California (Figure 1). The MPAs used for this study were implemented in northern California in 2012, north central California in 2010, central California in 2007 and southern California in 2003 and 2012. For the remainder of this study, we combined north central and northern California

sites due to limited sample sizes for these regions and similar biogeography (Griffith et al., 2016). Many MPAs had some form of protection implemented before the initiation of the MLPA, including all MPAs in the northern Channel Islands, although most MPAs saw increases in protection after MLPA (e.g. expanded boundaries or stricter regulations). Each MPA designation comes with varied forms and levels of activities and protections, which include State Marine Reserves (SMRs), State Marine Conservation Areas (SMCA), SMCA (No-Take) and Special Closure (Figure 1). SMRs are fully protected no-take MPAs, with exceptions only for scientific research. SMCA allow limited, regulated fishing, unless designated as No-Take, in which case all extraction is prohibited. Special Closure MPAs restrict access and boating activities. We removed three MPAs from the analysis (Vandenberg SMR, Point Lobos SMR and Big Creek SMR) because protection started before our study period (before 1984) and the regulations were not substantially changed by the MLPA. For the other MPAs, we used the most recent designation and area



FIGURE 1 Distribution of selected MPAs included in this study in mainland California and the Channel Islands. Designation colours correspond to the type of protected area which includes State Marine Conservation Areas (SMCA), No-Take State Marine Conservation Areas (SMCA No-Take), State Marine Reserves (SMR) and Special Closures.

size data. This study did not involve any fieldwork, and therefore no field permits or permissions were required.

In California, the two canopy-forming kelp species are giant kelp (*Macrocystis pyrifera*) and bull kelp (*Nereocystis luetkeana*). Giant kelp is the dominant surface canopy-forming species from the USA/Mexico international border to Half Moon Bay (~37.5°), north of which bull kelp becomes the dominant species. We used quarterly maps of giant kelp and bull kelp canopy coverage (1984–2022), the earliest available, derived from Landsat satellite imagery, to examine trends and variability in kelp coverage inside and outside MPAs. These maps, obtained from [KelpWatch.org](https://kelpwatch.org), were generated using a decision tree classifier and multiple endmember spectral unmixing (MESMA) to estimate the fraction of kelp coverage in each 30m×30m pixel across California. The methods have been validated for giant kelp (Bell et al., 2020, 2023) and bull kelp (Hamilton et al., 2020), but the dataset does not distinguish between species, and so we refer to all surface canopy-forming kelp as 'floating kelp'. Given the lack of comprehensive datasets (i.e. using standardized methods at large temporal and spatial scales) to track kelp forest dynamics, we used kelp canopy coverage as an indicator of the broader ecosystem dynamics. We narrowed our analyses to MPAs with at least 10 pixels of potential kelp coverage (9000m²), meaning at least 10 pixels each occupied at some point during the time series, which resulted in the inclusion of a total of 54 MPAs (supplemental Table S1).

2.2 | Identifying reference sites for before-after control-impact paired series analysis

We used a Before-After Control-Impact Paired Series (BACIPS) approach to test for the MPA implementation effect on kelp coverage. This approach involves comparing paired affected (Impact) and unaffected (Control) sites before and after some event, for example, MPA implementation (Rodríguez-Rodríguez & Martínez-Vega, 2022). Assessing the impact at each site relative to a paired control is particularly important when assessing effects on a dynamic species with rapid turnover rates such as floating kelp. One major challenge with BACI design is in selecting control sites. Ideally, in the absence of the intervention, both sites would exhibit similar trends and variability, allowing the paired-BACI design to isolate the effect of the intervention (Stevens, 2002; Stewart-Oaten et al., 1986; Wauchope et al., 2021). It is rare to have time series data of sufficient length to confirm that the control and impact sites have similar dynamics pre-intervention, and therefore environmental proxies (e.g. substratum type, depth, swell exposure, water temperature, nutrient availability) are instead used to identify control sites. The 20+ years of Landsat data preceding the 2014–2016 MHWs capture multiple ENSO events, offering a baseline for detecting divergence between paired sites (Huang et al., 2017).

We identified a paired reference site for each MPA using the Landsat kelp canopy time series (Bell et al., 2023), which enabled us to directly compare the dynamics of control and impact sites prior to

MPA implementation. Our general approach was to identify a contiguous region of kelp coverage outside of each MPA that exhibited similar kelp canopy dynamics to those within the MPA prior to MPA implementation. We first calculated the total area of kelp coverage within each MPA. Kelp coverage was defined as 30m Landsat pixels that contained kelp at any point during the time series from 1984 to 2022. We then implemented a sliding window technique to find a number of pixels of kelp coverage outside of the MPA (Figure 2) that matched the coverage inside the MPA. We constrained our reference site search to a 0.5×0.5-degree window (~55km) centered on the MPA so that the reference site would be in the same geographic area as the MPA site. The reference site regions needed to be contiguous, though they could include a split. For example, a single reference site could consist of the sum of kelp pixels from two adjacent sides of an MPA (Figure 2). This approach also helped accommodate coastal complexity and ensured we could define suitable reference sites when continuous stretches of kelp habitat were not available on just one side. Our approach is distinct in using a long-term Landsat time series to systematically identify spatially constrained, pre-MPA-reference sites, enabling consistent comparisons across a statewide MPA network. For each potential reference site, we plotted the annual time series of maximum kelp coverage (Figure S2). We calculated the Spearman correlation of the annual time series of kelp canopy inside the MPA and in the potential reference site for the years from 1984 to the most recent MPA implementation year. The optimal reference site was determined by selecting the reference time series with the highest correlation with the MPA time series in this pre-implementation period. This approach potentially ensures that the environmental conditions within the chosen reference site closely align with those within the MPA (assessed in the following section). We removed any MPAs from further analysis where the correlation between the MPA and the best potential paired reference site was <0.5.

2.3 | Environmental characteristics of paired MPA and reference sites

We assessed the similarity of potential environmental drivers of kelp dynamics between paired MPAs and reference sites to determine whether our reference site selection process yielded comparison pairs that were well matched in environmental characteristics. We examined mean annual maximum sea surface temperature (°C), depth (m), sea surface nitrate (μmolL⁻¹) and maximum significant wave height (m). Sea surface temperature was calculated using the mean daily surface temperature from the NOAA Coral Reef Watch SST data (5km resolution), available from 1985 to 2022 (NOAA Coral Reef Watch, 2018). Sea surface nitrate was derived from the daily temperature data following Snyder et al., 2020, and in northern California (north of San Francisco) a temperature to nitrate relationship was derived from a time series of CalCOFI data (Snyder et al., 2020). The northern California temperature to nitrate relationship closely resembled a published relationship (García-Reyes

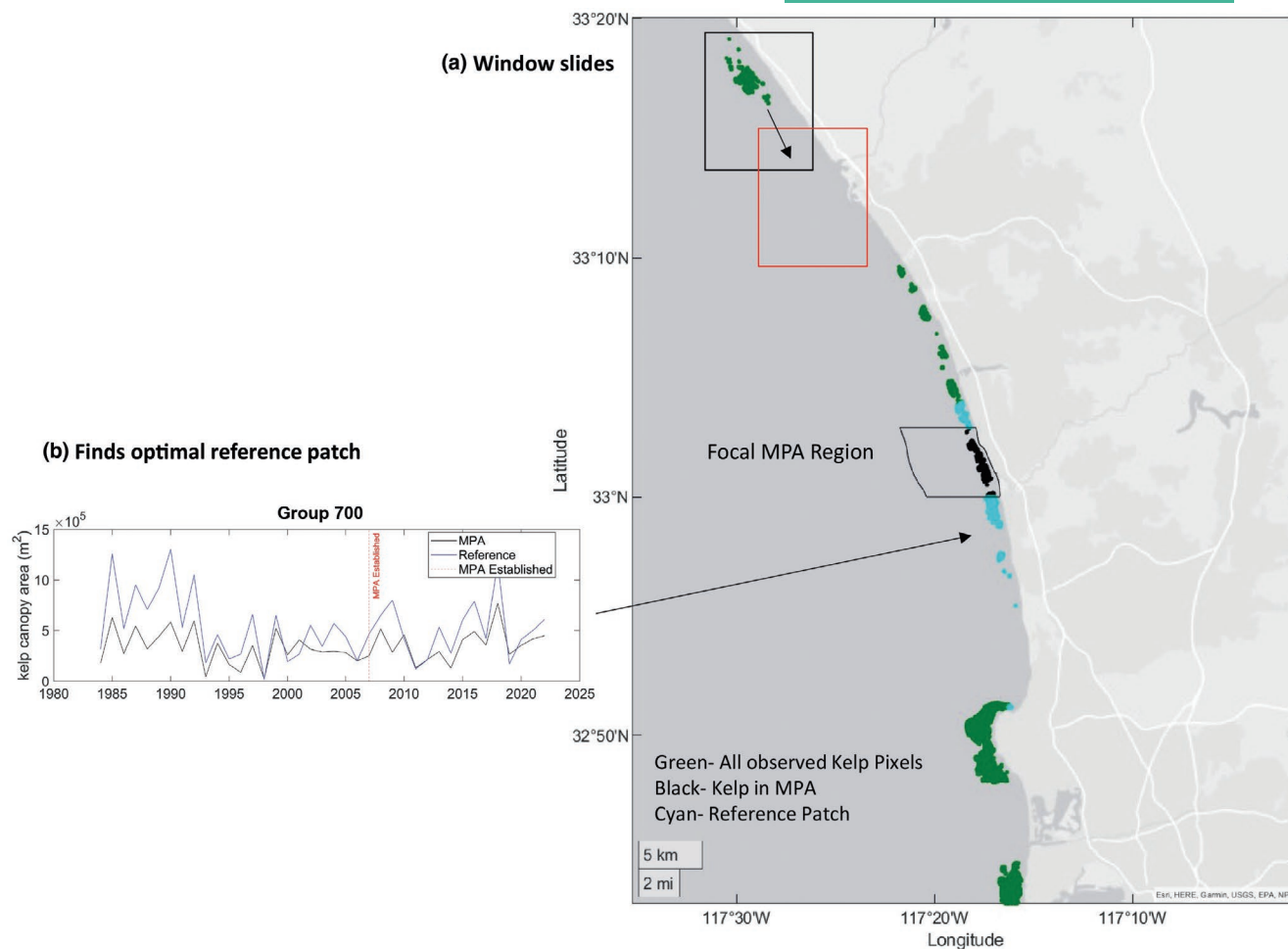


FIGURE 2 (a) Reference site selection method using a sliding window (0.5×0.5 -degree window centered on the MPA). The method identifies a patch of kelp canopy pixels outside the MPA that matches the size of the patch inside. The black polygon marks the initial search area, and the red box represents the sliding window moving along the coast to locate suitable patches. (b) Example of an annual time series comparing maximum kelp canopy between the MPA and a potential reference site. The sliding window identifies the reference patch where the annual dynamics of maximum kelp cover correlate most strongly with those in the MPA. Black pixels indicate the focal MPA region, and cyan pixels show the selected reference site after the search.

et al., 2014). Depth of each 30 m pixel was derived from the Seafloor Mapping project, except for areas around Port San Luis and to the far south near Imperial Beach, where data were obtained from NOAA National Centers for Environmental Information (NCEI) and from the General Bathymetric Chart of the Oceans (GEBCO) (www.gebco.net), respectively. Maximum significant wave height was obtained from the Coastal Data Information Program (CDIP) MOPv1.1 wave model on an hourly time scale at 1 km coastline segments (O'Reilly et al., 2016). All wave data prior to 2004 were hindcasted by developing a non-linear statistical model between CDIP data and data from one of 18 offshore US Army Corps Wave Information Study (WIS) model sites (1984–2019) (Bell et al., 2023). The site that produced the best model estimating CDIP wave height from WIS wave height, period, and direction was used to model the daily maximum significant wave height back to 1984 (Bell et al., 2023). For each variable we calculated the average of all grid cell values within the boundaries of each MPA and reference site. We then calculated the correlation between the MPA and reference site for

each environmental variable. We acknowledge a potential scale mismatch between some environmental variables and the kelp canopy data, due to the coarser resolution of the environmental datasets. However, this discrepancy was somewhat reduced when the data were aggregated to the extent of the MPAs.

2.4 | Changes in kelp canopy coverage after MPA implementation

To assess the effect of individual MPAs on kelp coverage, we conducted a BACIPS analysis for each MPA by estimating the difference between the maximum kelp coverage inside an MPA and the maximum coverage in its paired reference site in each year from 1984 to 2022. We then coded each year as before or after the implementation of the MPA. The coverage difference before implementation represents a baseline for kelp coverage, reflecting natural differences in kelp coverage between the MPA and the reference

site. A significant change in the differences after implementation reflects the effect of MPA protection relative to this baseline. We then scaled the MPA effect by subtracting the average difference pre-implementation from the difference post-implementation and then dividing that value by the average coverage of kelp within the MPA prior to implementation. We expressed the size of any MPA effect as a scaled impact (S_{imp}).

$$S_{imp} = \frac{\Delta \text{ postImplementation} - \Delta \text{ preImplementation}}{\text{Average Kelp preImplementation}} \quad (1)$$

The result is a change in kelp coverage between the sites, expressed as a fraction of the baseline level of kelp. Thus, an impact of 0.5 would indicate that kelp increased within the MPA relative to the reference site by 50% of the amount present at the MPA site prior to implementation. Similarly, negative values indicate decreases in kelp in the MPA relative to the reference.

We tested the significance of MPA effects on kelp within each individual MPA using a generalized linear model with Gaussian error, testing whether the average difference between an MPA and its reference site changed after implementation relative to before (treating implementation as a binary variable) (Table S1). All residuals were inspected for normality of error and autocorrelation. As a result of the large number of separate tests, we adjusted for multiple comparisons to control the family-wise error rate using a Holm adjustment (Holm, 1979).

We assessed whether MPAs collectively increase kelp coverage across the entire study region and regionally across northern ($n=12$), central ($n=10$) and southern California ($n=32$). Here we used linear mixed effects models (Bates et al., 2015), treating the identity of each MPA as a random effect to account for baseline variation in kelp coverage among MPAs and their paired reference sites. Our full model included a binary variable indicating whether the MPA had been implemented and the random effect of MPA identity. We compared this full model to the reduced model with only the random effect using a likelihood ratio test (ANOVA in R). A significant p -value from this test indicates that including MPA implementation as a factor significantly improved the model fit. All statistical analyses were performed in R version 4.3.2 (R Core Team, 2024).

2.5 | Resistance to and recovery from the 2014–2016 marine heatwaves in MPAs

We assessed whether MPAs mitigated the initial response to (i.e. loss) and supported the subsequent recovery of kelp coverage to the 2014–2016 MHWs. We define resistance and recovery as the relative kelp coverage during and after the MHWs, respectively, compared to a pre-heatwave baseline. To assess resistance and recovery, we calculated the proportional change in the mean annual maximum kelp coverage during the MHWs (2014–2016) and after them (2017–2022), relative to a 30-year baseline period (1984–2013) prior to the MHWs, for each MPA and its paired reference site (i).

$$\text{Resistance}_i = \frac{\text{mean}(\text{Kelp}_{i,\text{heatwaves}})}{\text{mean}(\text{Kelp}_{i,\text{baseline}})} \quad (2)$$

$$\text{Recovery}_i = \frac{\text{mean}(\text{Kelp}_{i,\text{recovery}})}{\text{mean}(\text{Kelp}_{i,\text{baseline}})} \quad (3)$$

We then used a paired t -test to compare the resistance and recovery metrics of kelp forests between MPAs and their paired reference sites, treating each MPA and its reference site as a pair. The test checks whether the mean difference in these metrics is significantly different from zero. We also performed these tests separately for each region (north, central and southern). We calculated recovery and resistance differences between MPAs and paired reference sites for each region individually and all regions combined.

3 | RESULTS

3.1 | Reference finder

We first identified paired reference sites for each MPA in our study area (supplemental Figure S1). We removed MPA-reference pairs where the best correlation between kelp dynamics before MPA implementation was <0.5 , leaving us with 54 MPA-reference pairs (Table S1). The mean correlation between kelp dynamics for MPA-reference pairs before MPA implementation was 0.79, and after MPA implementation, the mean correlation was 0.68. MPAs and their paired reference sites experienced similar temperature, nutrient and wave conditions ($r=0.99$, 0.99 and 0.73 , respectively; Figure 3a–d). Depth was less correlated between MPA and reference sites ($r=0.32$), with reference sites having greater depth on average (Figure 3b).

3.2 | Changes in kelp canopy coverage after MPA implementation

The outputs from the linear mixed effects model, which collectively assessed the differences between MPA and reference sites, indicated a significant overall positive effect of MPAs on kelp coverage ($\beta=13,212$, $SE=5454$, $t=1.966$, $p<0.05$) (Table S4). The magnitude of this effect was estimated at $13,212\text{ m}^2$ of kelp canopy, indicating an average expansion of canopy coverage inside MPAs compared to their reference sites. Prior to implementation, the MPAs in our study exhibited an average kelp canopy of approximately $150,000\text{ m}^2$, so the observed increase represents about 8.5% of the pre-implementation canopy. Although this effect size is substantial, it is relatively modest when considered in the context of the natural variability within the system. For instance, the average coefficient of variation for the yearly maximum kelp canopy at MPA sites before the impact was 108%, highlighting substantial fluctuations in kelp coverage due to other factors (Table S4). At regional scales, we found no significant effect

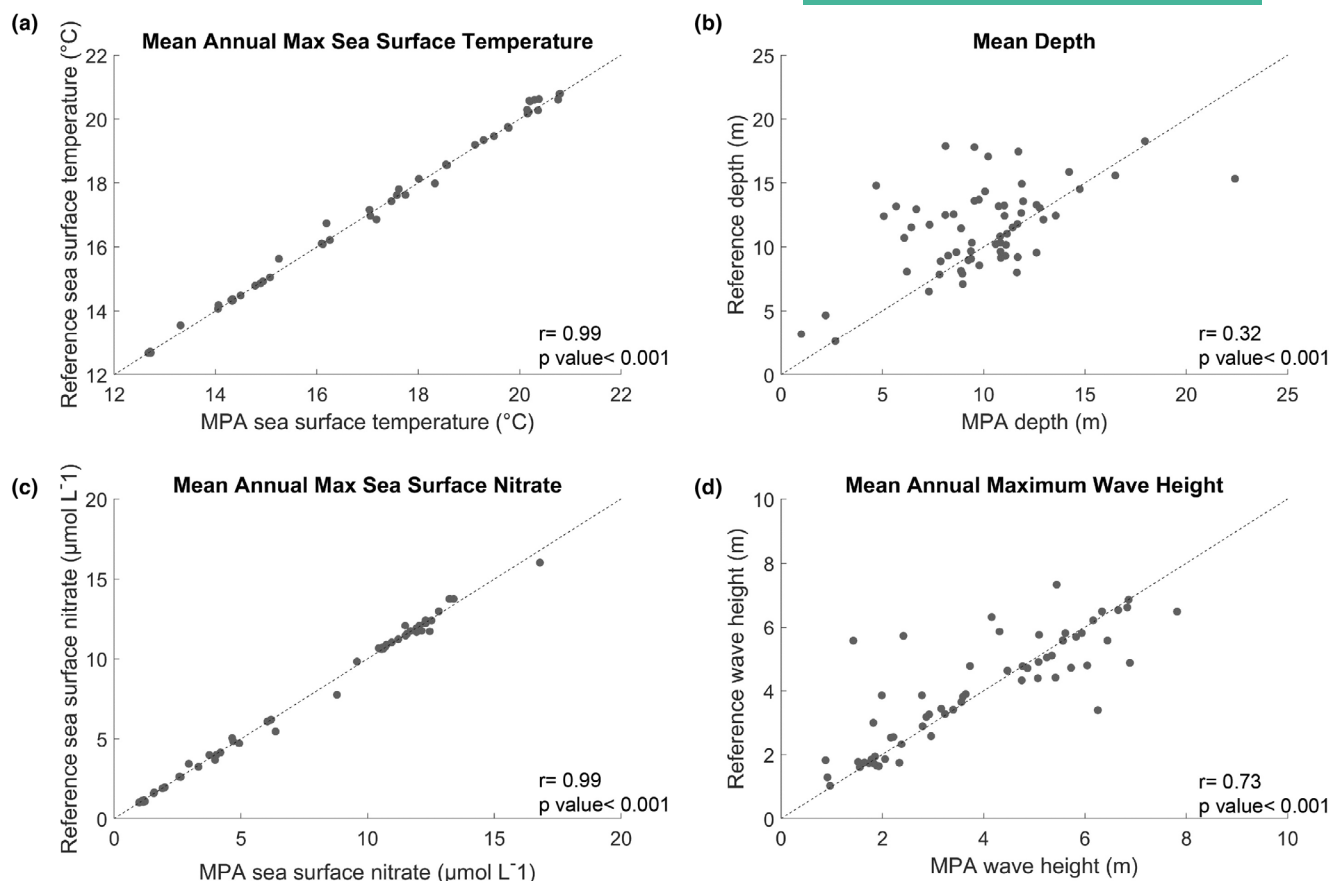


FIGURE 3 Mean annual maximums of sea surface temperature (a), mean depth (b), sea surface nitrate (c) and wave height (d) plotted against MPAs and their respective reference sites to measure the correlation of environmental variables between sites.

of MPA implementation across southern California ($\beta = 5228$, $SE = 3913$, $t = 1.336$, $p = 0.18$), no significant effect in central California ($\beta = 44,908$, $SE = 23,724$, $t = 1.893$, $p = 0.06$) and a significant negative effect in northern California ($\beta = -17,918$, $SE = 4797$, $t = -3.735$, $p < 0.01$) (Table S4).

BACIPS analysis indicated that changes in kelp canopy coverage after MPA implementation were highly variable among MPAs. When each of the 54 sites was assessed individually, rather than as a network, 6 MPAs showed a significant positive impact of MPA protection on kelp coverage following a Holm adjustment (Table S2). These positive effects were predominantly located in southern California (Point Vicente SMCA (No-Take), Point Conception SMR, Skunk Point SMR, Anacapa Island SMCA and Anacapa Island SMR), with one site from the central California region (Asilomar SMR) (Figure 4). One site showed a significantly negative impact, Saunders Reef SMCA, which is in northern California (Figure 4). No significant impact of MPA protection was observed for the remaining sites.

3.3 | Resistance to and recovery from the 2014–2016 MHWs

During the 2014–2016 MHWs, kelp coverage declined across most of California, including at the 54 sites assessed in this study (Figure 6c).

Statistically, we found no evidence that MPAs provided resistance to kelp during the MHWs when comparing paired MPA and reference sites (paired t -test; $p = 0.93$; Figures 5a and 6c). There were also no significant effects at the regional scale (northern California $p = 0.2$, central California $p = 0.33$, southern California $p = 0.84$). The median for the difference in MPA and reference relative coverage during the MHWs across all sites was 0.02 (0.09 for southern California, 0.01 for central California and 0.004 for northern California, Figure 5a). Across the 54 sites, 3 showed no change between MPA and reference, 30 exhibited more kelp in the MPA and 21 exhibited less kelp in the MPA.

Kelp recovery following the MHWs was more variable across California. While the central California and southern California regions showed the highest relative kelp coverage, the northern California region showed the lowest relative area (Figure 7a,b). Relative kelp coverage following the MHWs was significantly higher in MPA sites than in their reference sites across the entire study area (paired t -test; $p < 0.01$; Figure 5b) and southern California ($p < 0.01$; Figure 5b), but not for central California ($p = 0.67$) and northern California ($p = 0.38$). As a result, the statewide pattern is likely to be driven by southern California. The median for the difference in MPA and reference relative coverage after the MHWs across all sites was 0.1 (0.19 for southern California, 0.13 for central California and 0.01 for northern California, Figure 5b). Across the 54

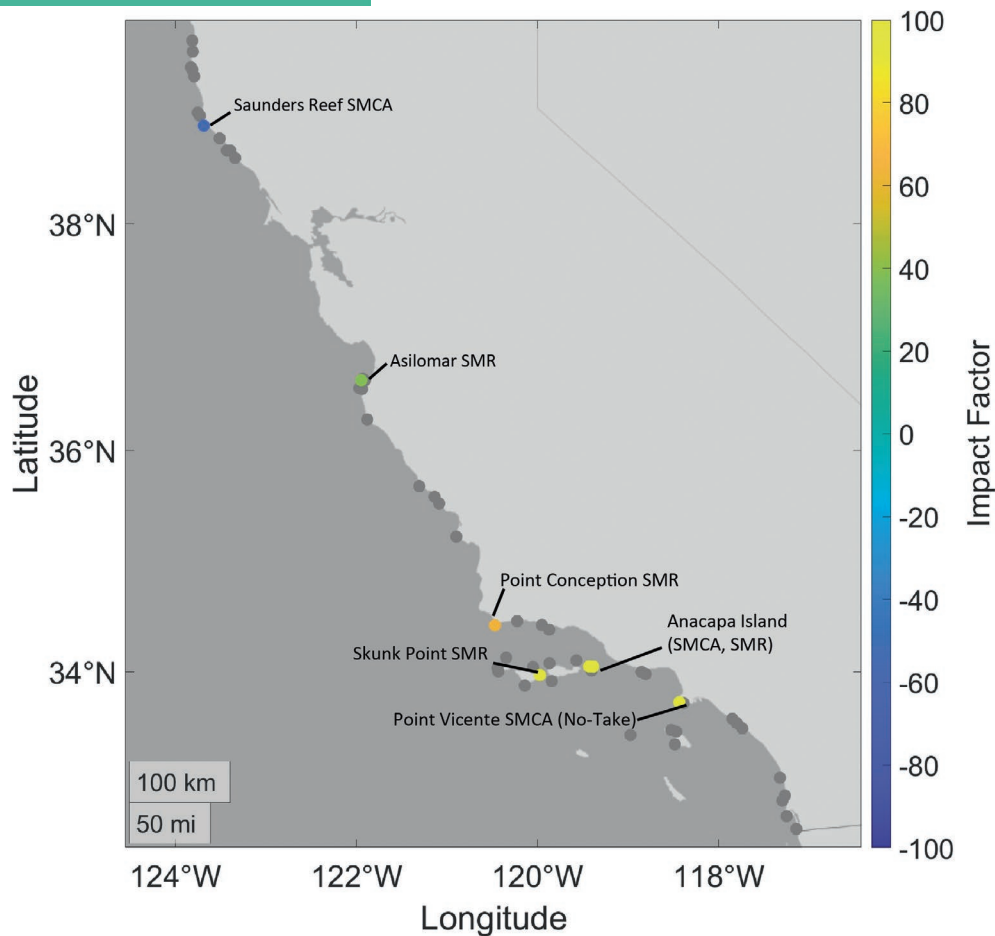


FIGURE 4 Map displaying the MPA impact factor on kelp forest coverage for each MPA. Labelled sites indicate those with a significant effect at the $p < 0.05$ level; the direction of the impact factor (positive or negative) is represented by the colour bar. Grey dots denote MPAs in the study that did not show a significant effect of protection. Anacapa Island State Marine Conservation Area (SMCA) and Anacapa Island State Marine Reserve (SMR) are two distinct MPAs located in close proximity.

sites tested for recovery, 1 showed no change, 38 exhibited greater kelp recovery in the MPA and 15 exhibited less.

4 | DISCUSSION

Testing the indirect effects of MPA protection on kelp is difficult and requires large amounts of data to distinguish MPA effects against the high natural variability in the abundance of these organisms, long-term data that are rarely available from in-water monitoring programmes. Furthermore, MPA effects are likely to be subtle and spatially variable when the mechanisms of effect are indirect (e.g. trophic or competitive) and the species of interest display high natural variability, as is the case with canopy-forming kelps (Rassweiler et al., 2021). We addressed these challenges by using a long-term and large-scale remote sensing dataset of kelp coverage to develop a method to systematically identify suitable reference sites for testing the effects of MPA protection on kelp coverage over the entire state of California. We found evidence that MPA protection can increase kelp coverage, but the effects were highly spatially

variable and, when they occurred, were weak. However, following a large disturbance to kelp, in the form of severe and intense MHWs, the positive effect of MPAs on recovery became more evident, with southern regions showing the most pronounced effect. This finding provides evidence that MPAs can provide climate resilience for kelp forests.

Globally, the effectiveness of MPAs in protecting kelp forests varies widely, with outcomes depending on the specific environmental pressures and management strategies in each region (Filbee-Dexter et al., 2024). However, characterizing this variability is challenging because of the difficulty of acquiring the data necessary to perform comprehensive assessments of indirect MPA effects. To isolate the effects of MPA protection, scientists need to carefully select reference sites to control for the many other factors that affect kelp coverage (Hamilton et al., 2010; Jainese et al., 2024). In many cases, not enough attention has been paid to the selection of reference sites. Often, the selection process relies on qualitative decisions about habitat similarity (Borja et al., 2012; Hopkins et al., 2016). Here, we provide a novel method to quantitatively identify reference sites based on temporal patterns of kelp variability before the

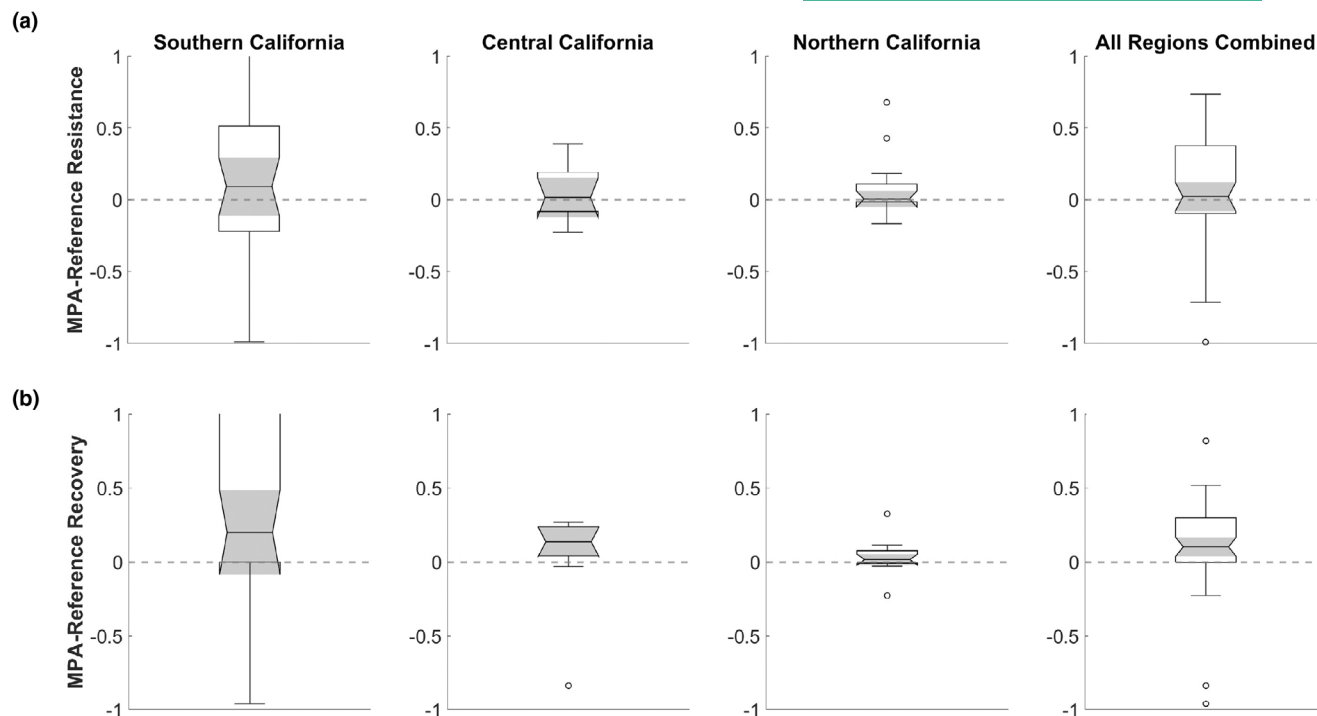


FIGURE 5 Difference between MPA and reference site relative coverage (a) resistance to (2014–2016) MHWs and (b) recovery (2017–2022) from the 2014–2016 MHWs across the three regions and all regions combined in California, relative to the pre-marine heatwave baseline (1984–2013). Resistance and recovery are measured as the difference of mean annual relative kelp coverage between MPAs and paired reference sites. A positive value indicates that MPAs had higher relative kelp coverage compared to the reference sites, while a negative value indicates that MPAs experienced lower relative kelp coverage compared to the reference sites. Values near zero indicate minimal differences in kelp coverage between MPAs and reference sites.

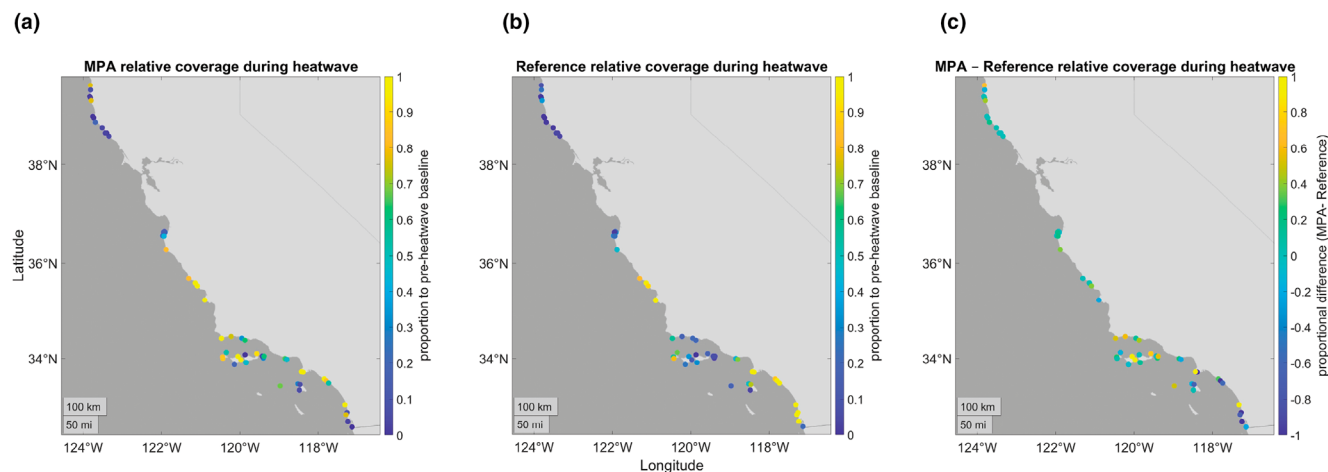


FIGURE 6 Relative kelp coverage during the 2014–2016 MHWs in MPAs (a) and reference sites (b) across California. Panel (c) shows the difference between MPA and reference kelp relative coverage during the MHWs. A value of 0 for relative coverage represents complete loss, while a value of 1 indicates complete recovery to pre-MHW values. If the difference of MPA–reference sites in relative coverage of kelp is 0, it means that the MPAs did not benefit kelp recovery following the MHWs, while a value of 1 indicates that MPAs fully supported kelp recovery relative to the reference sites. A value of –1 indicates that kelp recovery in MPAs was lower compared to the reference sites.

implementation of MPAs. The large spatial and temporal coverage of our satellite data enabled us to identify areas that responded similarly to environmental variability before MPA implementation, thus controlling for the effects of this variability.

In addition, the framework developed in this study for selecting reference sites offers a scalable solution that can be adapted to other regions and nations with floating kelps and other habitat-forming species that can be detected using remote sensing, making it a valuable

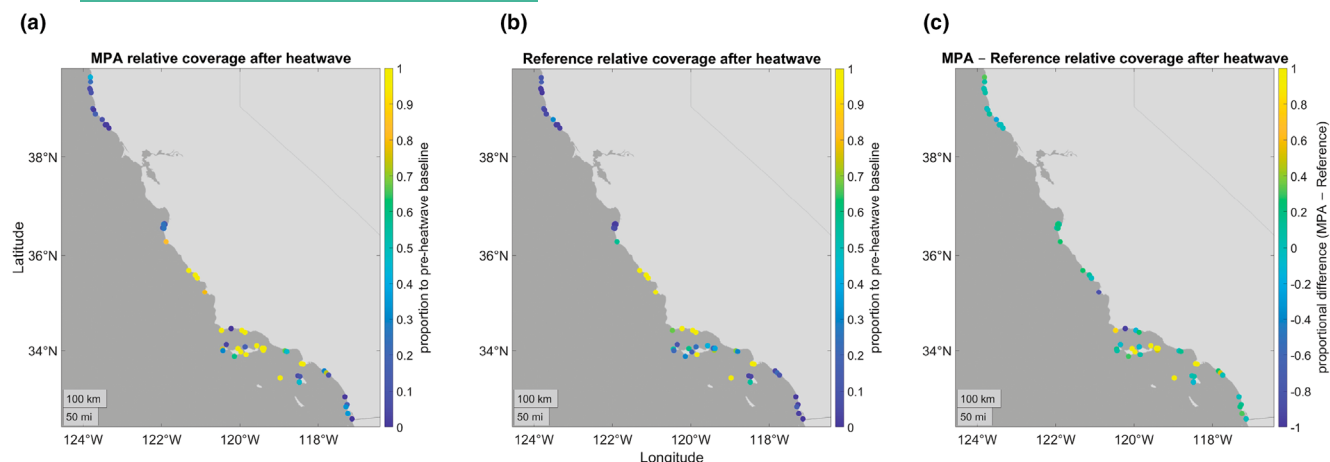


FIGURE 7 Relative kelp coverage after MHWs (2017–2022) for kelp coverage in MPAs (a) and reference sites (b) across California. Panel (c) illustrates the difference in relative kelp coverage between MPAs and reference sites after the MHWs. A negative value indicates that MPAs had lower relative kelp coverage compared to reference sites, while a positive value suggests that MPAs had higher relative kelp coverage than reference sites. Values near zero indicate minimal differences in kelp coverage between MPAs and reference sites.

tool for global MPA research and adaptive management efforts. Still, there are additional considerations associated with selecting reference sites that we did not address here. For example, MPAs are expected to affect adjacent control areas, either through the movement of fish from the MPA into the reference areas (i.e. spillover) or changes in fishing pressure in the surrounding regions (Di Lorenzo et al., 2016). The proximity of reference sites to MPAs may therefore increase the likelihood that the reference areas will be affected by MPAs. This may cause us to underestimate the effects of MPAs on marine populations and ecosystems in cases where reference sites are enhanced by MPAs (e.g. spillover) or overestimate the effects if reference sites are depleted, for example, by displaced fishing effort.

Our BACIPS analysis of the impacts of MPA implementation on kelp coverage produced mixed results. We found a positive impact of protection in 6 out of 54 sites, a negative impact in 1 and no significant effect in the remainder of the sites. One potential explanation for the lack of effect in most sites may be the high natural variability in kelp coverage that masked any MPA effect signal. It can be difficult to detect impacts using BACIPS approaches for species that experience large fluctuations in coverage through time, such as giant and bull kelp (Rassweiler et al., 2021). Furthermore, kelp forest community structure varies across regions, which has important implications for how MPAs influence trophic cascades, one mechanism by which MPAs may benefit kelp forests. Kelps and other primary producers are heavily influenced by the abundance of grazers; in California, these are primarily sea urchins. In southern California, two of the most important urchin predators are California sheephead (*Bodianus pulcher*) and California spiny lobster (*Panulirus interruptus*), both of which are fished. This could explain why 5 of the 6 MPAs with significant positive impacts of MPA protection were in southern California, which agrees with previous studies that found higher abundance of both species inside MPAs (Eisaguirre et al., 2020; Honeyman et al., 2023; Kumagai et al., 2024; Lenihan

et al., 2022; Selden et al., 2017; Smith et al., 2023). The effect of predation by sheephead and lobsters is not present in Central and northern California, where they are rare (Hopf et al., 2022; Rogers-Bennett & Catton, 2019). Here, sea stars and sea otters (in parts of central California) are the primary urchin predators, neither of which is fished. This could explain the pattern in northern California ($\beta = -17,918$), where an overall significant effect was seen after MPA implementation with a negative impact on kelp (Kumagai et al., 2024). Thus, the effect of MPAs on trophic cascades is expected to be stronger in southern California.

Across our entire dataset, MPAs lead to a significant increase in kelp coverage, but those coastline scale results masked substantial spatial variability. When broken down by region, MPAs in northern California had a significant negative effect on kelp, while MPAs in the central and southern regions had strong positive effects; effects that were not significant when analysed separately but were strong enough together to yield a significant positive result when the whole coastline is considered. These regional differences, along with the huge site-to-site variation we observed, suggest we should be cautious in extrapolating these results to new geographies. For example, while we have some understanding of how fishing pressure affects species densities, we lack a comprehensive understanding of how these changes in fished species across California influence the rate and speed of kelp forest responses at a broader scale. The time it takes for kelp forests in MPAs to recover may also be impacted by protected area size and proximity to anthropogenic stressors (i.e. population sizes, river mouths, water treatment plants) (Bhat, 2003; Brander et al., 2007; Kumagai et al., 2024; Lester & Halpern, 2008). Further research is needed to better understand the time frame within which conservation policies can effectively yield results, given the regional variability in kelp forest food webs, as well as the influence of environmental conditions, fishing pressure and MPA protection level.

Despite weak evidence for an overall effect of MPA protection on kelp coverage over the long term, we did find stronger evidence for positive effects of protection following the 2014–2016 MHWs. The combined disturbance of MHWs, which were associated with declines in kelp across the state (Bell et al., 2023) and the loss of sea stars, an important urchin predator, may have created a situation where indirect MPA effects on kelp coverage were easier to detect. For example, increased California sheephead and lobster abundance in MPAs may have reduced the likelihood of large urchin barrens forming after kelp declines from MHWs in southern California. In Norway, a study provided evidence of this phenomenon, where the recovery of kelp became more apparent in regions where environmental conditions favoured the proliferation of urchin predators. This helped facilitate a reverse shift from barrens to kelp (Christie et al., 2019). Therefore, the observed increase in relative kelp coverage after the MHWs could be related to the concurrent higher abundances of urchin predators inside MPAs, which appear to be size-dependent and influenced by reduced fishing pressure (Eisaguirre et al., 2020; Selden et al., 2017). These results support the use of MPAs as a climate resilience management tool, reinforcing their expansion not just to limit fishing pressure but to sustain key species that buffer ecosystems during MHWs.

The effect of MPA protection on the recovery of kelp coverage after the MHWs was higher in southern regions, where fished urchin predators are protected within MPAs. There was no clear effect of MPA protection on recovery after the MHWs in the central California and northern California regions, where the greatest declines of kelp were documented following the loss of seastars and many areas transitioned to an urchin barren state. Only sea otters remain as a major urchin predator (in central California) and this species is not directly impacted by MPAs because it is already a protected species and is not a targeted fishing species (Eisaguirre et al., 2020; Nichols et al., 2015). Our findings align with a recent study in southern and central California, which showed that trophic cascades enhance climate resilience, but only when urchin predators are fished species that benefit from MPAs (Kumagai et al., 2024). In northern California, there was a regional outbreak of purple urchins, which led to large declines in kelp abundance inside and outside of MPAs (McPherson et al., 2021; Rogers-Bennett & Catton, 2019). The absence of urchin predators in northern California may limit the effectiveness of MPAs in offering climate resilience to marine heatwave impacts, highlighting how MPA outcomes can vary depending on biogeographic conditions (Kumagai et al., 2024). Another factor is that MPAs were implemented at different times across the regions, and thus, when the MHWs occurred, they had been protected for a different number of years. Therefore, there may have been regional differences in the degree to which predators had recovered before the MHWs, leading to differences in the strength of trophic cascades.

5 | CONCLUSION

Our findings revealed a small positive effect of MPA protection on kelp coverage, although outcomes varied substantially across individual

MPAs. We conclude that while MPA protection benefits help in some locations, the magnitude of this benefit is relatively small and not generalizable across MPAs in California. However, positive effects became clearer post-MHWs, especially where MPAs protect urchin predators and maintain trophic cascades, as seen in southern California. This result is unsurprising, as the MPA network in California spans a large biogeographic range. Our results add to a growing body of research confirming the strong spatial variability in MPA effectiveness across the network (Arafeh-Dalmau et al., 2023; Honeyman et al., 2023; Kumagai et al., 2024; Smith et al., 2023; White et al., 2025; Ziegler et al., 2023). As nations move forward to effectively protect 30% of the oceans by 2030 while adapting to climate change (Ma, 2023), we provide empirical evidence of the potential of MPAs to enhance climate resilience, which can help set realistic expectations for MPA outcomes under climate impacts. Kelp forest ecosystems are ecologically and economically important for coastal communities, making it essential to understand how protection influences kelp dynamics and response to climate impacts, particularly as climate change and fishing pressure pose ongoing challenges to kelp forests.

MPAs can enhance climate resilience to MHWs impacts for kelp forest ecosystems in some cases, yet more research is needed on the underlying mechanisms and environmental conditions that drive these outcomes. Here, we suggest the importance of trophic cascades, and more research to better understand the ubiquity of this mechanism globally is warranted. In addition, factors such as MPA traits, environmental conditions and anthropogenic stressors have all been shown to contribute to MPA performance, and their effects across large networks warrant more study. Given the projected increase in climate change-related disturbances (IPCC, 2021) including intensifying MHW exposure for kelp forests (Arafeh-Dalmau et al., 2025), it is likely that protection will need to be supplemented with additional climate adaptation strategies to support the conservation and recovery of ecologically important species such as kelp forests.

AUTHOR CONTRIBUTIONS

Conceptualization, Emelly M. Ortiz-Villa and Kyle C. Cavanaugh; methodology, Emelly M. Ortiz-Villa, Andrew Rassweiler and Kyle C. Cavanaugh; software, Emelly M. Ortiz-Villa and Kyle C. Cavanaugh; validation, Emelly M. Ortiz-Villa and Kyle C. Cavanaugh; formal analysis, Emelly M. Ortiz-Villa, Andrew Rassweiler and Kyle C. Cavanaugh; investigation, Emelly M. Ortiz-Villa and Kyle C. Cavanaugh; resources, Kyle C. Cavanaugh; data curation, Kyle C. Cavanaugh and Tom W. Bell; writing original draft preparation, Emelly M. Ortiz-Villa and Kyle C. Cavanaugh; writing, review and editing, Emelly M. Ortiz-Villa, Kyle C. Cavanaugh, Andrew Rassweiler, Nur Arafeh-Dalmau, Jennifer E. Caselle and Katherine C. Cavanaugh; visualization, Emelly M. Ortiz-Villa; supervision, Kyle C. Cavanaugh; project administration, Kyle C. Cavanaugh; funding acquisition, Kyle C. Cavanaugh.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

Data available from the Environmental Data Initiative Digital Repository: <https://doi.org/10.6073/pasta/93b47266b20bc1782c8df9c36169e372> (Bell et al., 2022).

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Table S1. MPA site names, region, designation, and year of implementation. Designation acronyms correspond to the type of protected area which include State Marine Conservation Areas (SMCA), No-Take State Marine Conservation Areas (SMCA No-Take), State Marine Reserves (SMR), and Special Closures.

Table S2. Scaled Impact results with corresponding impact *p* value adjusted for yearly max average. MPAs in bold represent sites with a significant *p*-value. MPAs highlighted in green show significant *p*-value sites corresponding to a positive scaled impact factor, while MPAs highlighted in red show a negative impact factor or positive scaled impact.

Table S3. Summary of *t*-test results and median differences for recovery and resistance metrics across different regions (All Regions, South, Central, and North). Columns include the significance test result (*h*, where 1 indicates rejection of the null hypothesis), *p*-values, confidence interval (CI) bounds, *t*-statistic values, degrees of freedom (*df*), and the observed median differences. Significant

results ($p < 0.05$) were observed for recovery metrics in "All Regions" and "South," indicating a meaningful difference in recovery. Resistance metrics did not show significant differences in any region.

Table S4. Results of impact estimates and statistical analyses across regions (All Regions, Southern California, Central California, and Northern California). Columns include the impact estimate, standard error (SE), t -value, intercept variance, degrees of freedom (df), chi-square values, and p -values. Significant findings ($p < 0.05$) were observed in "All Regions" and "Northern California," with Northern California showing a negative impact estimate and the strongest significance ($p = 0.00023$). Results for Southern and Central California were not statistically significant, with p -values of 0.1817 and 0.05897, respectively.

Figure S1. An example of a panel illustrating the results for a Marine Protected Area (MPA) and its corresponding reference site, as identified by the reference finder method. Green pixels represent floating kelp canopy area derived from Landsat imagery. Red bounding boxes outline the MPA shapefile. Magenta pixels indicate the kelp habitat within the MPA of interest, while cyan pixels denote the selected reference site. The R value represents the correlation between the time series of kelp habitat area for the MPA and reference site prior to MPA implementation. "Habitat area" refers to the kelp canopy area within the magenta-highlighted MPA.

Figure S2. Example of a panel presenting the time series for a Marine Protected Area (MPA) (black line) and its corresponding reference site patch (blue line). A red marker indicates the implementation date for the MPA. Below the MPA name in the title, the R value is displayed, representing the correlation between the MPA and reference habitat time series after MPA implementation.

Figure S3. Example of a timeseries of deltas created for each site, illustrating the differences between MPA and REF. This analysis examines whether the delta (i.e., the difference between MPA and REF) significantly changes during the post-MPA period (impact = 1) compared to the pre-MPA period. The model treats site ID as a random variable, allowing each site to have its own average difference.

Figure S4. Scaled impact per MPA displayed geographically from north to south (arranged left to right). Red bars represent MPAs with significant p -values and negative scaled impact factors, while green bars indicate MPAs with significant p -values and positive scaled impact factors.

Figure S5. We calculated the coefficient of variation (CV) of annual kelp canopy coverage at each site, comparing the periods before and after MPA establishment. This analysis was used to assess whether MPA implementation dampened interannual variability in kelp coverage. We found no significant difference in CV before versus after MPA implementation (paired t -test; $p = 0.499$), suggesting no evidence of reduced variability in kelp canopy following protection.

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